Bukang-liwayway v3.3

Illustrated printing, assembly and stringing guide

Use with the written Printing & Assembly Guide (parts list, print settings, tuning and care) and stl/HARDWARE.md for the size you print.

!! SAFETY: ALWAYS WEAR EYE PROTECTION

Wear safety glasses whenever you build or finish the parts: snipping supports and necks, drilling, trimming and cutting line can flick sharp bits of plastic into your eyes. The insert iron runs at about 240 °C: keep it in its stand, never leave it on alone. Print, set inserts and glue in fresh air. Keep cutters, the iron and loose screws away from small children.

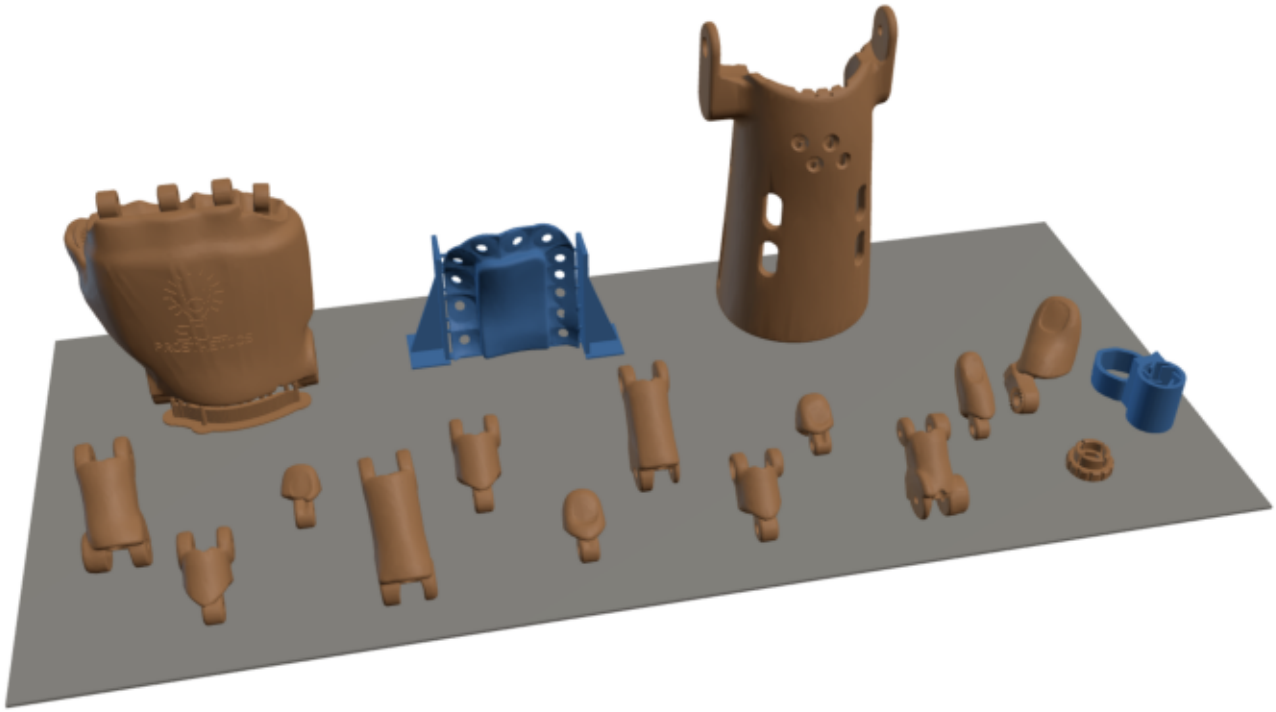


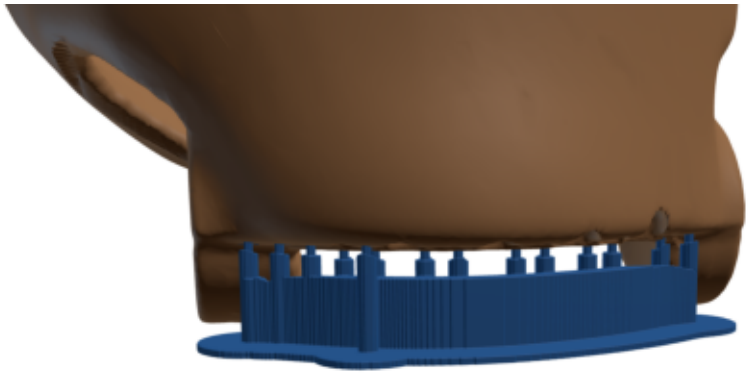


Order: print · 1 inserts · 2 fingers · 3 knuckle axle · 4 little finger · 5 thumb + knob · 6 return cords
7 flexor lines · 8 wrist · 9 tensioner · 10 grip pads · 11 palm cover · 12 straps and liner

Printing — every part in its print direction

Load each STL as it is (never rotate or scale it). PETG = tan, TPU = blue. 0.4 mm nozzle, 0.2 mm layers.





palm_back_print.stl: the blue snip-off support holds the rim on 14 thin necks between the line-tunnel exits. Snip the necks flush after printing.

Loading the blocker

Creality Print / Orca / PrusaSlicer: drag both files in together, answer Yes to "multiple parts", right-click the blocker → Change type → Support blocker.

Cura: turn off "drop models to build plate" and "keep models apart", load both, blocker → Per Model Settings → Don't support overlaps.

Back of the hand

Print palm_back_print.stl standing on its wrist end. Normal supports ON, and load its support blocker file (..._support_blocker.stl) with it, so no support goes into a hole or line tunnel. 5 mm brim.

Fingers, thumb, knob

Fingers and thumb stand at 45°, knuckle end down, nail side up. Supports on the BUILD PLATE ONLY: that keeps every support out of the screw holes. Brim. The thumb knob needs no supports.

TPU parts

Palm cover: palm_flap_print.stl (fins and sole bar built in). Art holder (optional): art-holder.stl, standing on its back end, no supports. TPU 95A, slow 20–30 mm/s, 4 walls, little retraction.

Screw holes

Never need support: they all print lying sideways and are small (2.6–3.4 mm). Afterwards run a 3.5 mm drill through the 3.4 mm holes by hand, never the 2.6 mm pilots.

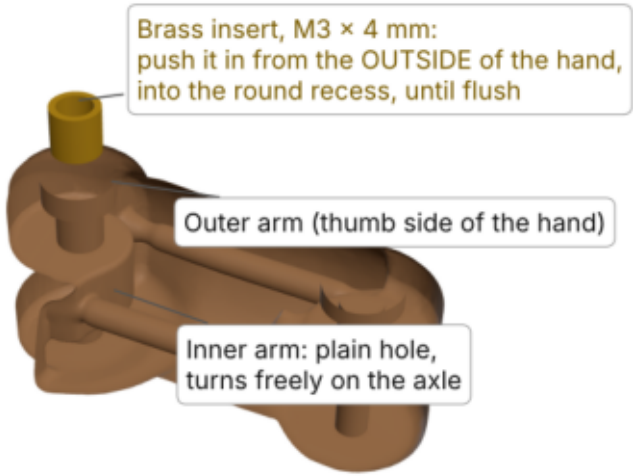
1 Heat-set inserts — set all 16 first

M3 brass, 4.2 mm across × 4 mm long. Soldering iron at about 240 °C for PETG; let its weight push the insert in, press flush with something flat and cool.

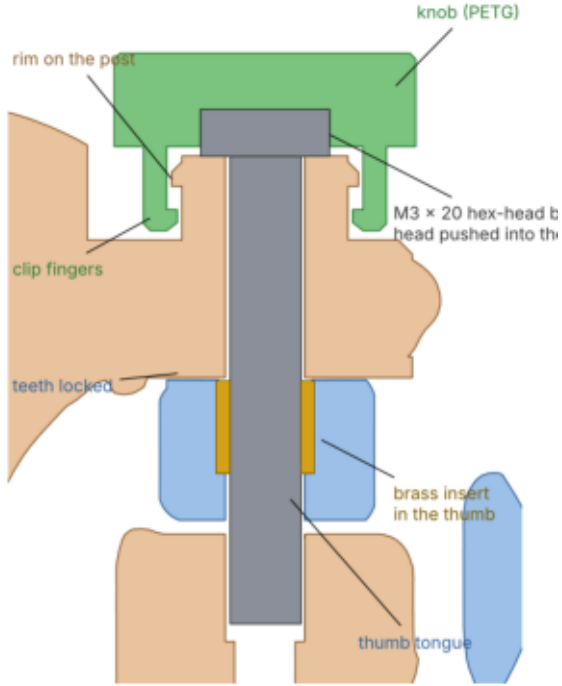
12 in the cover step (red rings)



1 in the index knuckle segment, outer arm



1 in the thumb tongue (gold)



Not in the palm at the thumb

The palm has a plain hole there: the knob's bolt passes through it into the insert in the thumb.

2 in the wrist ear pads

One in each round ear pad of the back of the hand, where the cuff arms meet it. Push in from outside, flush.

Every insert

Goes in from OUTSIDE and finishes flush. The index one sits right under the axle head; the thumb one goes in from the face with the ring of teeth. One cover hole near the thumb-side wrist corner can also take a 5 mm insert.

Parts and screws

Stainless M3 from a hardware store. No nuts. Tighten every joint only until it swings freely with no side wobble.

Screw / part	Qty	Where
M3 × 45 button head	1	Knuckle axle: index side → index, middle, ring knuckles → insert in the index outer arm
M3 × 12 button head	8	Middle + tip joints of index, middle, ring; little finger knuckle + tip joint
M3 × 20 hex head (DIN 933)	1	Thumb knob bolt: head pressed into the knob, threads into the thumb insert
M3 × 6 button head	2	Wrist hinge: cuff arm → insert in the ear pad (blue threadlocker)
M3 × 5 flat head (countersunk)	12	Palm cover, all round its edge, into the step inserts
M3 × 4 flat head (countersunk)	4	Line clamps on top of the cuff
M3 heat-set insert 4.2 × 4 mm	16	Cover step 12, wrist ears 2, index outer arm 1, thumb 1
Blue threadlocker (removable)	—	Knuckle axle and the 2 wrist screws

2 Build each finger

Join 1 to 2, then 2 to 3. Each joint is a barrel in a cup held by one M3 × 12 button head, put in from the side with the round recess so the head sits flush. It threads into the plastic.

3 Knuckle axle

Line up the index, middle and ring knuckles on their tongues. Blue threadlocker on the M3 × 45, in from the index side through all three, into the insert in the index OUTER arm. Tight, the index still swings freely.

4 Little finger

Its own M3 × 12 from the recessed side.

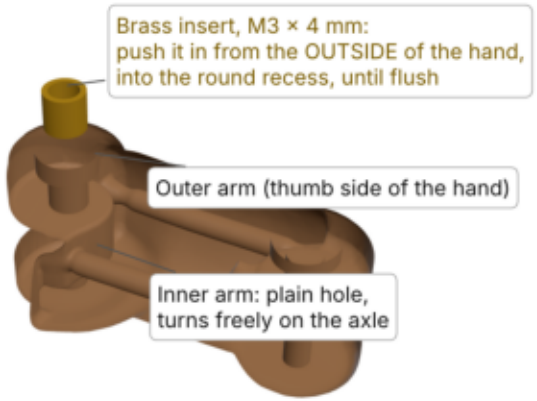
5 Thumb + knob

See the thumb knob page: hex bolt into the knob, tongue in its slot (teeth toward the knob), knob clicked over its rim, tighten by hand.

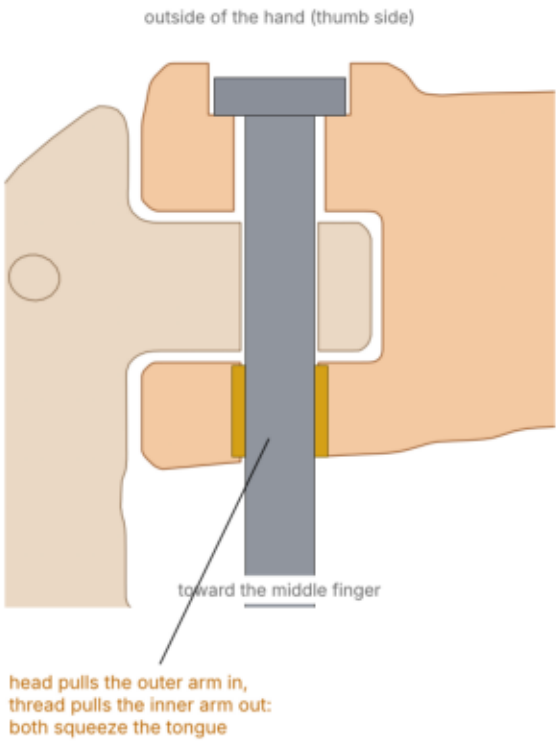
Bukang-liwayway v3.3 — index knuckle: where the heat-set insert goes now

The insert moved to the OUTER arm, right under the head of the M3 × 45 axle. Head and thread now grip the same arm, so tightening the axle can no longer squeeze the finger onto the palm.

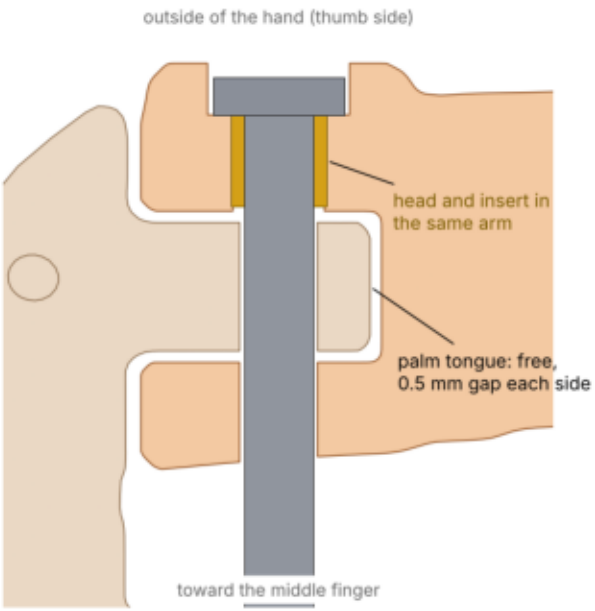
Index knuckle segment (finger-index-1)



v3.2: thread in the INNER arm



v3.3: thread in the OUTER arm



Put it together

1 Heat-set the insert into the OUTER arm of the index knuckle segment, from outside, flush with the bottom of the round recess (iron about 240 °C).
2 Line up the index, middle and ring knuckles on their palm tongues.
3 Push the M3 × 45 in from the index side, through all three knuckles, and screw it into the insert until the head sits in the recess.

Check

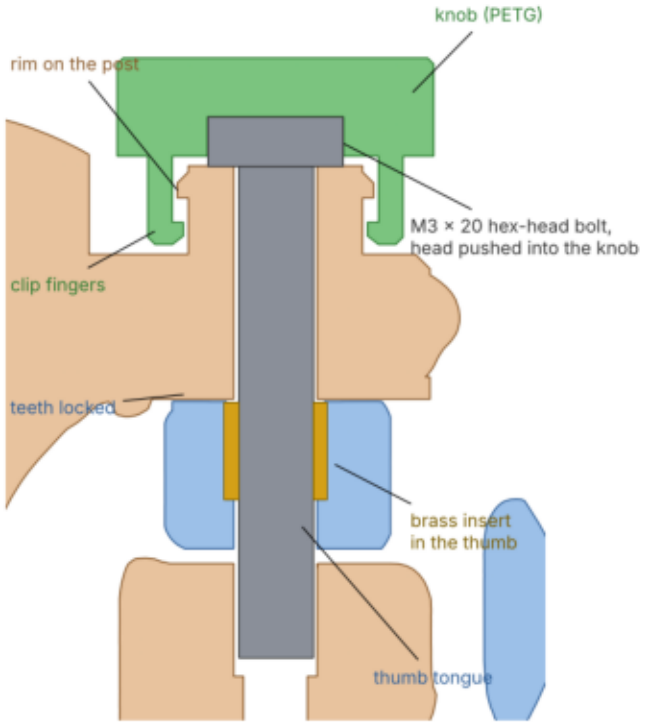
- Tighten it fully: the index finger must still swing freely. Nothing gets clamped now.
- A drop of blue threadlocker on the thread keeps it from working loose.

Bukang-liwayway v3.3: tool-free thumb knob

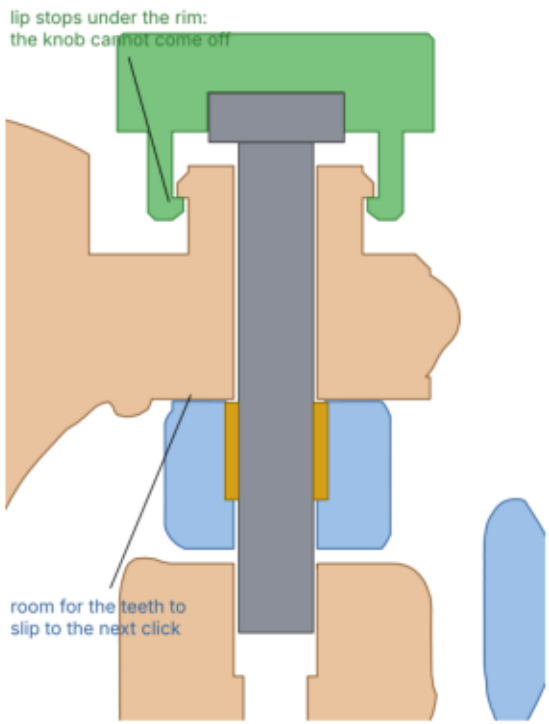
Hold the thumb, turn the knob back about 2 turns, swing the thumb to the next click, turn the knob tight. No screwdriver.

The knob clips over a rim on the hand, so it can only lift 1 mm and can't come off by itself.

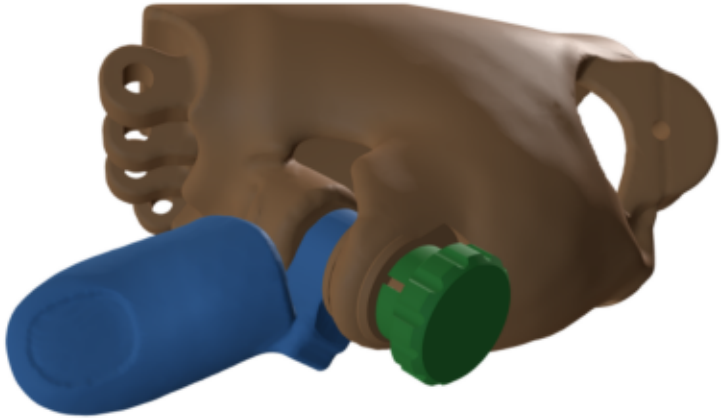
Locked: knob tight



Loosened to the stop (1 mm)



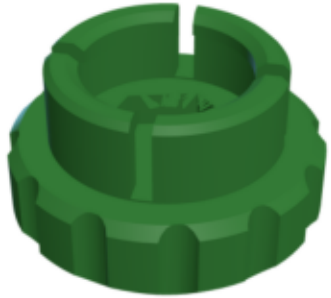
On the hand



Knob, underside



Print: top down, no supports



Put it together

1 Push the M3 × 20 hex head up into the knob's hex pocket until it bottoms (warm the head with the iron if it is tight).

2 Thumb insert set as before. Line up the thumb in its slot, push the bolt through, and turn the knob to start the thread.

3 Press the knob down until its fingers click over the rim, then tighten it by hand.

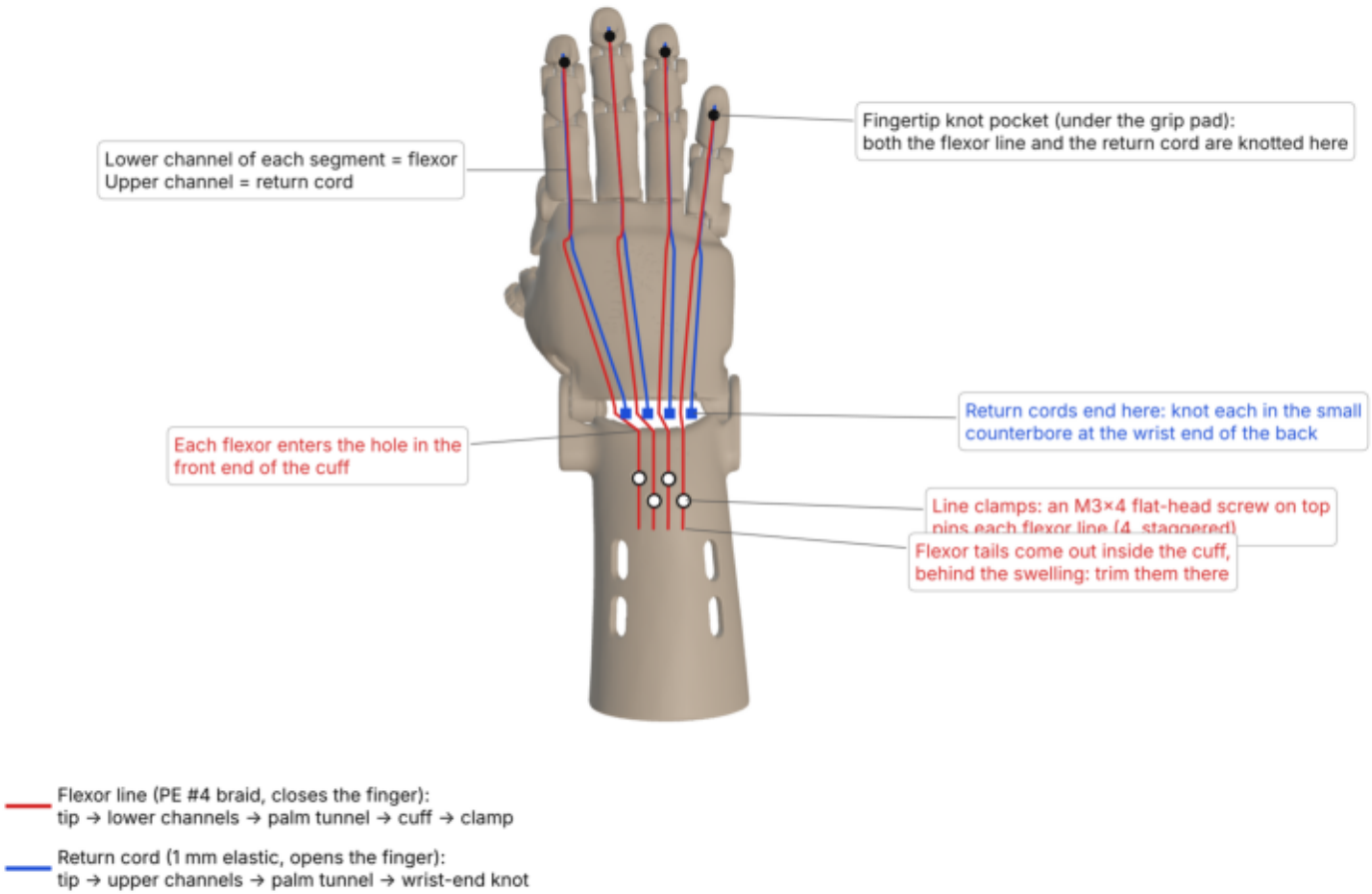
Changing the thumb position

Hold the thumb, turn the knob back until it stops (about 2 turns), swing the thumb to the next click (about 20° apart), then tighten the knob firmly by hand. Never force it past the stop.

Taking it apart

Turn the knob back to the stop, ease the four fingers over the rim with a small flat screwdriver while turning, then unscrew.

Stringing — the four fingers



6 Return cords (blue)

Knot the 1 mm elastic in the fingertip knot pocket (a surgeon's knot and a drop of glue). Thread it through the UPPER channel of each segment, then its tunnel in the back of the hand. Knot it in the small counterbore at the wrist end, just snug enough to hold the finger open.

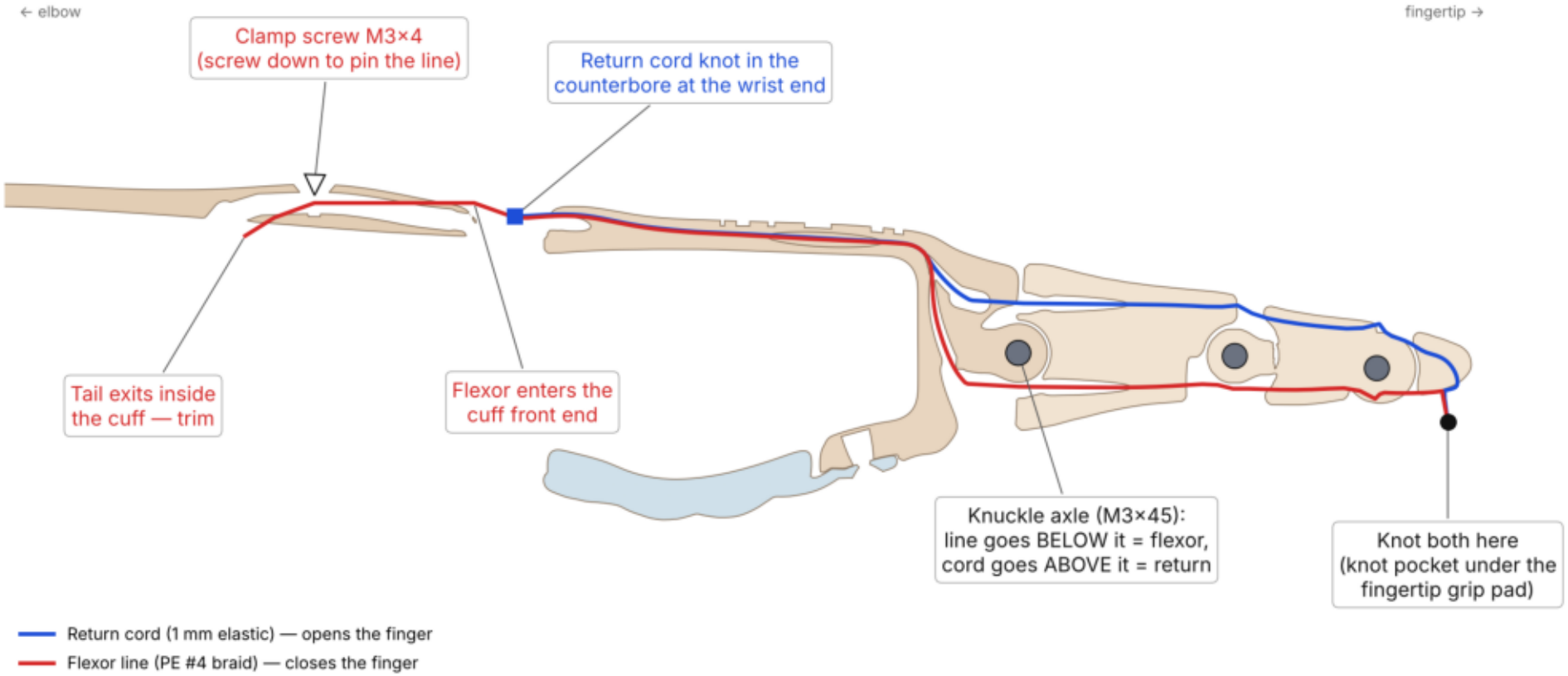
7 Flexor lines (red)

Knot the PE #4 braided line in the same fingertip pocket and seal the knot. Thread it through the LOWER channel of each segment, into the knuckle block and through its tunnel to the wrist end. Holes are 2–2.5 mm with 3 mm funnel mouths; a thin wire or long needle helps.

Tip

Do one finger completely before the next, and label the line ends with tape so you know which line belongs to which finger.

Stringing — side view of one finger (middle finger, cut lengthwise; the others are the same)



Which channel is which

Flexor = below the joint screws (palm side). Return = above them (back of the hand). Pulling the flexor bends every joint; the elastic springs them straight.

8 Wrist

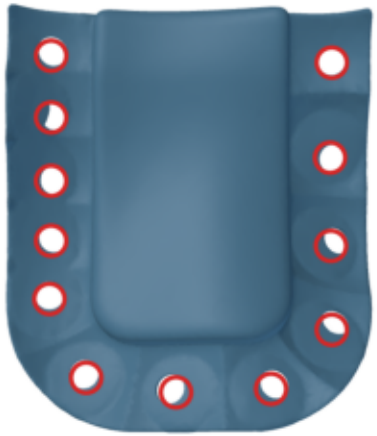
Join the back of the hand to the cuff with two M3×6 screws through the cuff's hinge arms into the brass inserts in the ear pads. Each screw is the hinge pin too: a drop of blue threadlocker, and tighten only until the cuff swings freely.

9 Tensioner

Feed each flexor into its hole in the front end of the cuff. With the wrist straight and fingers open, pull snug and screw down its M3×4 clamp until flush. Trim the tail inside the cuff. Bend the wrist down; adjust until all four fingers close together.

Grip pads, palm cover, straps

Palm cover (TPU), inside face



Back of the hand, palm side: the step



Red rings: the 12 cover screws (M3 × 5 flat-head into brass inserts), all round the edge. No snaps, no glue.

10 Grip pads

Inner-tube pads into the 5 fingertip and thumb recesses. Sand both dull, wipe with alcohol, thin film of E6000 or contact cement on both, wait until tacky (2–5 min), press, clamp 24 h. Cure 72 h before a child wears it.

11 Palm cover

Lay the cover in its step. Start all 12 M3×5 flat-head screws loosely into the inserts, then tighten each until its head sits flush in the TPU. Don't overtighten. No glue: unscrew the 12 to change a cover.

12 Straps and liner

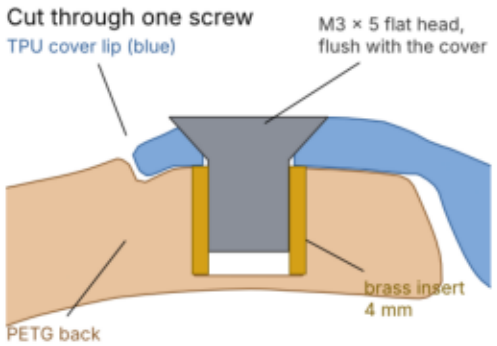
Sew the two straps on through the cuff slots and sand any edge that touches skin. Liner: a thin cotton sock or stockinette on the child's hand and forearm. The hand slides in through the wrist end like a glove; the cover stays on.

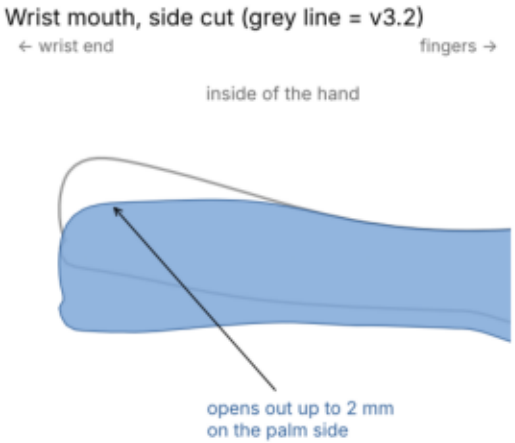
Bukang-liwayway v3.3: palm cover held by 12 screws into brass inserts

No snaps and no glue: unscrew 12 screws to take the TPU cover off and swap it. The wrist end of the hand opens out a little so the hand slides in like a glove.

Back of the hand, palm side up: 12 brass inserts in the step round the opening







Screws: 12 × M3 × 5 flat head (countersunk), stainless.
Inserts: 12 × M3 brass heat-set, 4.2 mm across × 4 mm long
(a 5 mm insert also fits at the thumb-side wrist corner).
The cover edge strip is now 9.4 mm wide all round
(was 6 mm except on the little-finger edge).

Put it together

1 Set the 12 inserts into the step round the opening, from outside, flush (iron about 240 °C).

2 Slide the hand in through the wrist end, then lay the TPU cover in its step.

3 Start all 12 screws loosely, then tighten them only until each head sits flush in the TPU. Over-tightening squashes the TPU.

Changing the cover

Unscrew the 12 screws, lift the old cover off, lay the new one in and screw it back. The inserts stay in the hand; nothing is glued. No snap pegs, so the cover prints with supports on the build plate only where the slicer asks for them.

Bukang-liwayway v3.3: art holder

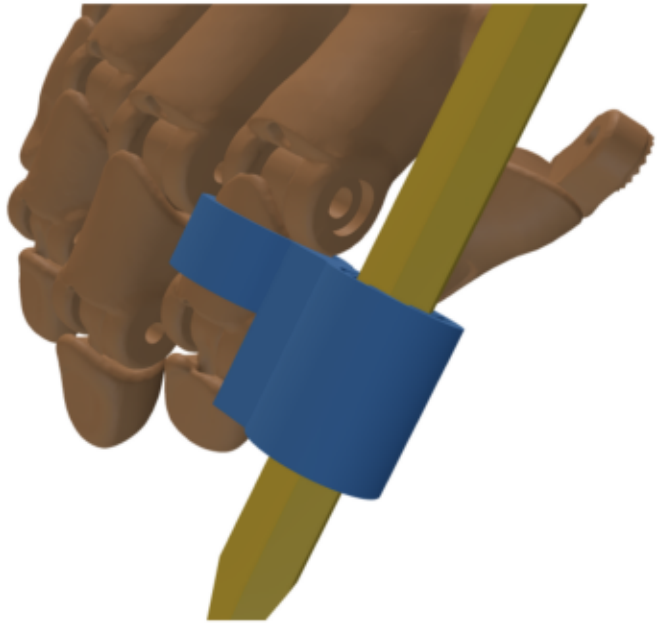
One TPU part, no screws. Slide it over the index fingertip onto the middle segment and push a pencil, pen, crayon, marker or brush into the tube.

Thin fins inside, all leaning the same way, fold over as the tool goes in and hold it: about 4.5 mm (thin brush) to 11.5 mm (fat marker).

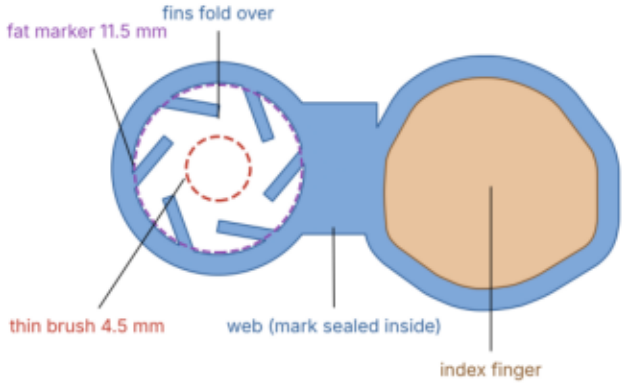
On the hand



Close up



Cut across the finger



As printed (standing on its back end)



Print

TPU 95A, standing on its back end (as it comes in the file), no supports. Slow (20–30 mm/s) for the thin fins.

Use

Slide it over the index fingertip, tube on the thumb side, onto the middle segment. Push the tool in from the back, point forward, with a little twist.

Set the thumb to its pinch click: in the more open clicks it meets the tube when the fingers close.